

# SAMORAH

## Business Requirements Document

### v3.0 | Enterprise Custom Platform

Luxury Handmade Scented Candles — Complete Build Reference

This document is the single source of truth for every Claude prompt session

|                  |                                                                   |
|------------------|-------------------------------------------------------------------|
| Platform         | Next.js 15 + TypeScript + Supabase                                |
| Frontend         | Next.js 15 + Tailwind CSS + Shadcn UI + Framer Motion             |
| Backend          | Supabase — PostgreSQL + Auth + Storage + Edge Functions + pg_cron |
| Payments         | Razorpay — UPI, RuPay, Cards, Wallets, Net Banking, EMI, COD      |
| Shipping         | Shiprocket API — rate calc, AWB, NDR, tracking, returns           |
| Email/Marketing  | Resend + React Email — 15 templates + sequences                   |
| Hosting          | Vercel (Frontend) + Supabase Cloud (Backend)                      |
| Monthly Cost     | ₹65–500/month at launch                                           |
| Document Version | v3.0 — June 2025 (Updated: Added 14 missing sections)             |

HOW TO USE THIS BRD WITH CLAUDE:

- Step 1: Upload this entire BRD and say: 'Read it and acknowledge. Do not write code yet.'
- Step 2: Start each phase with: 'Claude, referring to SAMORAH BRD, execute Phase [N].'
- Step 3: Build one file at a time. Never ask Claude to build entire pages in one prompt.
- Step 4: Reference section numbers (e.g., 'Section 8.3') when asking for specific features.

# 1. Project Overview

|                     |                                                                                 |
|---------------------|---------------------------------------------------------------------------------|
| <b>Brand Name</b>   | Samorah                                                                         |
| <b>Industry</b>     | Luxury Handmade Scented Candles                                                 |
| <b>Platform</b>     | Custom-built — Next.js 15 + Supabase (No WooCommerce / No Shopify)              |
| <b>Domain</b>       | Purchased via GoDaddy — connected to Vercel, auto SSL via Let's Encrypt         |
| <b>Database</b>     | Supabase PostgreSQL — with pg_cron for scheduled jobs (abandoned cart, backups) |
| <b>Image CDN</b>    | Cloudinary — auto WebP/AVIF, CDN, transformation                                |
| <b>Payment</b>      | Razorpay — UPI, RuPay, Cards, Net Banking, Wallets, EMI, COD                    |
| <b>Shipping</b>     | Shiprocket API — NDR management included                                        |
| <b>Email</b>        | Resend + React Email — 15+ templates + marketing sequences                      |
| <b>Monthly Cost</b> | ₹65–500/month at launch vs ₹2,500–8,000/month on Shopify                        |

## 1.1 Core Objectives

- Editorial luxury ecommerce experience — Trudon, Aesop, Byredo, Dior aesthetic quality
- Complete Admin Panel: products, orders, inventory, SKUs, coupons, blog, analytics, banners
- Full inventory management with SKUs, low-stock alerts, audit trail, CSV bulk operations
- RBAC: 5 user roles — Customer, Editor, Manager, Admin, Super Admin
- Razorpay: UPI, RuPay, Cards, Wallets, EMI, COD — with idempotent webhook handling
- Shiprocket: AWB, tracking, NDR management, COD reconciliation
- GST-compliant Tax Invoices with HSN codes — legally required in India
- Marketing automation: abandoned cart sequences, welcome flow, festival campaigns
- Loyalty system, referral program, gift cards — Phase 2 ready schema included
- Performance: <2s load, Lighthouse >90, Supabase pg\_cron for background jobs

## 2. Technology Stack

### 2.1 Frontend

| Layer         | Technology              | Purpose                                                          |
|---------------|-------------------------|------------------------------------------------------------------|
| Framework     | Next.js 15 (App Router) | SSR, ISR, SSG, Server Components, API Routes                     |
| Language      | TypeScript              | Type safety — all components, stores, API routes typed           |
| Styling       | Tailwind CSS            | Utility-first — no CSS bloat, full design control                |
| UI Components | Shadcn UI               | Accessible headless components — themed to Samorah brand         |
| Animations    | Framer Motion + GSAP    | Luxury micro-animations, scroll effects, page transitions        |
| State         | Zustand (8 stores)      | Cart, auth, wishlist, checkout, UI, search, notifications        |
| Forms         | React Hook Form + Zod   | Client + server validation, TypeScript-safe schemas              |
| Data Fetching | TanStack Query          | Server state caching, background refetch, stale-while-revalidate |
| Admin Tables  | TanStack Table          | Sortable, filterable, paginated admin data tables                |

### 2.2 Backend & Infrastructure

| Service        | Technology              | Purpose                                             |
|----------------|-------------------------|-----------------------------------------------------|
| Database       | Supabase PostgreSQL     | Primary store — products, orders, users, inventory  |
| Auth           | Supabase Auth           | JWT, OAuth, OTP, Magic Link, email verification     |
| Storage        | Supabase Storage        | Backup storage — Cloudinary is primary image CDN    |
| Edge Functions | Supabase Edge Functions | Webhook handlers, payment verify, Shiprocket sync   |
| Scheduled Jobs | Supabase pg_cron        | Abandoned cart sweeps, backup exports, stock alerts |

|            |                               |                                                        |
|------------|-------------------------------|--------------------------------------------------------|
| Realtime   | Supabase Realtime             | Live order status in admin — new orders ping dashboard |
| RLS        | PostgreSQL Row-Level Security | Users read/write only their own data                   |
| API Routes | Next.js API Routes            | Razorpay webhook, Shiprocket sync, idempotent handlers |

## 2.3 External Integrations

| Service   | Provider                         | Purpose                                                                 |
|-----------|----------------------------------|-------------------------------------------------------------------------|
| Payments  | Razorpay                         | UPI, RuPay, Cards, Wallets, Net Banking, EMI, COD — idempotent webhooks |
| Shipping  | Shiprocket                       | Order sync, AWB, pickup, NDR management, tracking, returns              |
| Email     | Resend + Brevo (optional)        | Transactional + marketing automation sequences                          |
| Images    | Cloudinary                       | Auto WebP/AVIF, resize, CDN — 25GB free tier                            |
| CMS       | Sanity CMS                       | Blog, journal, editorial sections — editor-friendly                     |
| Analytics | GA4 + Meta Pixel + Clarity       | Traffic, conversion funnels, heatmaps, GA4 events                       |
| Search    | PG Full-Text (P1) / Algolia (P2) | Product search, autocomplete, AI semantic search (P2)                   |
| GST / Tax | Manual (HSN + GST rate fields)   | Legally required for Indian tax invoices                                |

## 3. Folder & Route Architecture

### 3.1 Folder Structure

| Directory                | Purpose                                                                                     |
|--------------------------|---------------------------------------------------------------------------------------------|
| src/app/(store)/         | Customer-facing storefront routes — home, shop, product, collection, cart, checkout         |
| src/app/(admin)/         | Admin panel routes — protected by Next.js Middleware (role: editor+)                        |
| src/app/api/             | Next.js API routes — Razorpay webhook (idempotent), Shiprocket sync, abandoned cart trigger |
| src/components/ui/       | Shadcn design system: Button, Input, Card, Modal, Toast, Drawer, Accordion, Tabs            |
| src/components/sections/ | Page sections: Hero, CollectionSlider, ScentExperience, BundleBuilder, Newsletter           |
| src/components/layout/   | Header, MegaMenu, Footer, AdminSidebar, AdminTopbar, AnnouncementBar                        |
| src/store/               | 8 Zustand stores — see Section 7 for full spec + hydration fix                              |
| src/hooks/               | useStore (hydration-safe), useCart, useWishlist, useAuth, useSearch, useToast               |
| src/lib/                 | supabaseClient.ts, razorpay.ts, shiprocket.ts, cloudinary.ts, resend.ts                     |
| src/services/            | productService, orderService, authService, inventoryService, couponService, loyaltyService  |
| src/types/               | Product, Variant, Order, User, Cart, Coupon, GiftCard, LoyaltyPoints, AuditLog              |
| src/config/              | constants.ts, featureFlags.ts, gst.ts (HSN codes + rates)                                   |
| src/styles/              | globals.css — CSS design tokens (all --var declarations from Section 4)                     |
| middleware.ts            | JWT check + role enforcement on every /account/* and /admin/* request                       |

### 3.2 Key Routes

| Route        | Rendering | Notes                                                                |
|--------------|-----------|----------------------------------------------------------------------|
| /            | ISR 60s   | Homepage — banner manager controls hero/announcement/campaign blocks |
| /shop/[slug] | SSG       | Product page — 12 sections                                           |

|                          |            |                                                                      |
|--------------------------|------------|----------------------------------------------------------------------|
| /collections/[slug]      | SSG        | Collection editorial page                                            |
| /bundles                 | CSR        | Interactive bundle builder                                           |
| /checkout                | SSR        | GST-aware — calculates CGST/SGST/IGST at runtime                     |
| /admin/inventory         | Admin only | SKU management, stock adjustments, CSV import                        |
| /admin/banners           | Admin only | Homepage Banner Manager — hero, announcement, campaign, footer quote |
| /admin/loyalty           | Admin only | Loyalty points, tiers, referrals                                     |
| /admin/gift-cards        | Admin only | Gift card generation and tracking                                    |
| /admin/ndrs              | Admin only | Shiprocket NDR (Non-Delivery Report) management                      |
| /api/razorpay/webhook    | API Route  | Idempotent — checks order.status before processing                   |
| /api/cron/abandoned-cart | API Route  | Called by Supabase pg_cron every hour                                |

## 4. Branding & Design System

### 4.1 Reference Brands

| Brand      | Reference & Takeaway                                                                  |
|------------|---------------------------------------------------------------------------------------|
| Trudon     | trudon.com — dark editorial, cinematic chapter layout, mega menu with campaign images |
| Aesop      | aesop.com — restrained luxury, high whitespace, editorial typography, no noise        |
| Byredo     | byredo.com — minimal black/white, no-noise luxury, product-first                      |
| Dior       | dior.com — cinematic campaign energy, collection editorial layout                     |
| Boy Smells | boysmells.com — modern candle UX, variant selector, fragrance notes UI                |
| Voluspa    | voluspa.com — lifestyle grid, atmospheric imagery, Instagram-feel gallery             |

### 4.2 CSS Design Tokens (globals.css)

| CSS Variable     | Value   | Usage                                   |
|------------------|---------|-----------------------------------------|
| --bg-primary     | #F5F2ED | Main site background — warm ivory       |
| --bg-secondary   | #ECE7E0 | Section backgrounds                     |
| --text-primary   | #181818 | Headings, primary text                  |
| --text-secondary | #5F5A55 | Body text, descriptions                 |
| --accent         | #9C826B | CTAs, highlights, prices, badges        |
| --border-soft    | #DDD6CD | Borders, dividers                       |
| --dark-bg        | #181818 | Dark sections, scent experience, footer |

### 4.3 Typography

| Element    | Font               | Specs                                 |
|------------|--------------------|---------------------------------------|
| H1 Hero    | Cormorant Garamond | Bold, 64–96px, letter-spacing -0.02em |
| H2 Section | Cormorant Garamond | SemiBold, 36–48px                     |
| Body       | DM Sans            | Regular 400, 16px, line-height 1.7    |

|        |                       |                                           |
|--------|-----------------------|-------------------------------------------|
| Labels | DM Sans               | Uppercase, letter-spacing 0.12em, 11–12px |
| Prices | Cormorant<br>Garamond | Medium 500, 20–24px                       |

## 4.4 Animation System

- Fade-in on scroll: opacity 0→1, translateY(20px)→0, 0.4s ease-out
- Product image hover: scale(1.03), 0.4s — never aggressive
- Navigation: transparent → warm ivory on scroll (0.3s)
- All animations respect prefers-reduced-motion media query



## 5. Website Architecture & Pages

### 5.1 Full Sitemap

| Route               | Page           | Key Sections                                                                                                                 |
|---------------------|----------------|------------------------------------------------------------------------------------------------------------------------------|
| / (home)            | Home           | Hero (Banner Manager), Signature Chapters, Brand Story, Scent Exp, Featured, Testimonial, Instagram Grid, Newsletter, Footer |
| /shop               | Shop All       | Filters (price, category, mood, fragrance family), grid, sort, pagination                                                    |
| /shop/[slug]        | Product Page   | 12 sections — gallery, variants, story, fragrance notes, reviews, recommendations                                            |
| /collections/[slug] | Collection     | Editorial hero, featured product, supporting product grid                                                                    |
| /bundles            | Bundle Builder | 3-candle composition builder — vessel filter, composition tray, savings display                                              |
| /cart               | Cart           | Items, quantities, coupon, gift note, gift card redeem, shipping estimate                                                    |
| /checkout           | Checkout       | Address → Delivery → Payment (Razorpay) → Review → Confirmation (GST invoice)                                                |
| /account/*          | Account        | Profile, addresses, orders, wishlist, loyalty points, reviews                                                                |
| /journal/[slug]     | Blog           | Editorial blog posts via Sanity CMS                                                                                          |
| /contact            | Contact        | Minimal form, FAQs, WhatsApp link                                                                                            |
| /archive            | Archive        | Retired fragrances grouped by volume                                                                                         |
| /gift-cards         | Gift Cards     | Purchase and redeem Samorah gift cards                                                                                       |

### 5.2 Homepage — Banner Manager Controlled Sections

**IMPORTANT:** All homepage editorial sections below are controlled by the Homepage Banner Manager (Section 27).

Editors can update hero heading, announcement text, campaign images, and newsletter quote **WITHOUT** touching code.

This eliminates the need for developer involvement for routine content updates.

| # | Section          | Controlled By                                 |
|---|------------------|-----------------------------------------------|
| 1 | Announcement Bar | Banner Manager — text, link, background color |

|    |                           |                                                                      |
|----|---------------------------|----------------------------------------------------------------------|
| 2  | Navigation + Mega Menu    | Code — structure fixed, campaign image per chapter is Banner Manager |
| 3  | Hero                      | Banner Manager — heading, subtext, CTA text/link, background image   |
| 4  | Signature Chapters Slider | Admin — collection management, cover images                          |
| 5  | Brand Story Block         | Banner Manager — image, heading, copy, CTA                           |
| 6  | Scent Experience (dark)   | Code — fragrance families from database                              |
| 7  | Air Chapter + Bundle      | Banner Manager — campaign images, CTA links for both blocks          |
| 8  | Featured Product          | Admin — mark product as is_featured = true                           |
| 9  | Testimonial Quote         | Banner Manager — quote text, attribution                             |
| 10 | Instagram Atmosphere Grid | Admin — Instagram Gallery Manager (Section 27)                       |
| 11 | Newsletter                | Banner Manager — headline, subtext                                   |
| 12 | Footer                    | Banner Manager — poetic footer statement                             |

### 5.3 Cart — Gift Features

- Gift Note field: free-text, max 200 chars — printed on packing slip and shown in order
- Gift recipient name (optional — for luxury gift-wrap orders)
- Occasion dropdown: Birthday | Anniversary | Diwali | Christmas | Just Because | Other
- Gift card redemption: apply gift card code (validates balance, deducts from total)
- Gift wrap option (Phase 2): checkbox for branded Samorah gift box (+₹99)

## 6. Authentication & Authorization

### 6.1 Authentication Methods

| Method           | Provider               | Notes                                                                                                                                                                        |
|------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Email + Password | Supabase Auth          | Email verification required, bcrypt hashing                                                                                                                                  |
| Google OAuth     | Supabase Auth + Google | One-click login, profile auto-fill                                                                                                                                           |
| OTP (Phone)      | Supabase Auth + SMS    | 6-digit OTP, 10-min expiry. <i>"Strict IP-based rate limiting on the /api/auth/otp request endpoint (e.g., max 3 requests per 15 minutes) to prevent SMS bombing fraud"?</i> |
| Magic Link       | Supabase Auth          | Passwordless email — 15-min expiry                                                                                                                                           |
| Guest Checkout   | N/A                    | No account needed — tracked by email + order number                                                                                                                          |

### 6.2 JWT Token Architecture

- Access Token: 15-minute expiry — stored in memory ONLY (never localStorage)
- Refresh Token: 7-day expiry — httpOnly Secure SameSite=Strict cookie
- Token rotation on every use — prevents replay attacks
- Next.js Middleware validates JWT on every /account/\* and /admin/\* request

### 6.3 RBAC — 5 Roles & Permission Matrix

| Action                | Customer | Editor | Manager | Admin | SuperAdmin |
|-----------------------|----------|--------|---------|-------|------------|
| Browse / shop         | Yes      | Yes    | Yes     | Yes   | Yes        |
| Write reviews         | Yes      | Yes    | Yes     | Yes   | Yes        |
| Publish blog posts    | No       | Yes    | Yes     | Yes   | Yes        |
| Edit homepage banners | No       | Yes    | Yes     | Yes   | Yes        |
| Add/edit products     | No       | No     | Yes     | Yes   | Yes        |
| Manage inventory/SKUs | No       | No     | Yes     | Yes   | Yes        |
| Process orders + NDR  | No       | No     | Yes     | Yes   | Yes        |
| Create coupons        | No       | No     | Yes     | Yes   | Yes        |
| Initiate refunds      | No       | No     | No      | Yes   | Yes        |

|                           |    |    |    |     |     |
|---------------------------|----|----|----|-----|-----|
| Manage loyalty/gift cards | No | No | No | Yes | Yes |
| Change user roles         | No | No | No | No  | Yes |
| Access audit logs         | No | No | No | No  | Yes |
| System settings/API keys  | No | No | No | No  | Yes |

# 7. State Management — Zustand

## 7.1 Critical: Next.js Hydration Fix

⚠ CRITICAL TECHNICAL REQUIREMENT — Must implement before any store uses localStorage persistence.

The Problem: Next.js renders pages on the server first. localStorage does not exist on the server.  
Using Zustand with persist(localStorage) causes a Hydration Mismatch Error — the site CRASHES on load.

The Fix: Implement a custom useStore hook that delays rendering persisted Zustand state until the client-side component has mounted. Pattern:

```
const useStore = <T, F>(store: (callback: (state: T) => unknown) => unknown, callback: (state: T) => F) => {
  const result = store(callback) as F;
  const [data, setData] = useState<F>();
  useEffect(() => { setData(result); }, [result]);
  return data;
};
```

Use this hook everywhere persisted store state is rendered in a component.  
Claude Prompt Note: When building any Zustand store with localStorage, reference this fix.

## 7.2 The 8 Zustand Stores

| Store            | State Fields                                                                    | Key Actions                                                                        |
|------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| useUserStore     | user, isLoggedIn, role, addresses, loyaltyPoints, tier                          | setUser, clearUser, updateProfile, addAddress, updateLoyaltyPoints                 |
| useCartStore     | items[], totalItems, totalPrice, coupon, giftCard, discount, giftNote, occasion | addItem, removeItem, updateQty, applyCoupon, applyGiftCard, clearCart, setGiftNote |
| useWishlistStore | items[], count                                                                  | addToWishlist, removeFromWishlist, syncToServer, isWishlisted                      |

|                      |                                                                       |                                                                     |
|----------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------|
| useCheckoutStore     | address, delivery, payment, gstBreakdown, step, orderId, isProcessing | setAddress, setDelivery, setGSTBreakdown, nextStep, prevStep, reset |
| useProductStore      | filters{price,category,mood,fragrance}, sort, view, searchQuery, page | setFilter, clearFilters, setSort, setView, setSearch, setPage       |
| useSearchStore       | query, results[], isOpen, isLoading, history[], aiSuggestions[]       | setQuery, setResults, openSearch, clearHistory, setAISuggestions    |
| useUIStore           | isMiniCartOpen, isMenuOpen, toasts[], activeModal, bannerData         | openMiniCart, toggleMenu, addToast, removeToast, openModal          |
| useNotificationStore | notifications[], unreadCount                                          | addNotification, markRead, markAllRead, clearAll                    |

## 7.3 Cart Persistence Logic

| Scenario           | Behavior                                                                    |
|--------------------|-----------------------------------------------------------------------------|
| Guest adds to cart | Zustand + localStorage — no login required, survives page refresh           |
| Guest logs in      | Guest cart merged with server cart — duplicates have quantities combined    |
| Checkout initiated | 15-minute stock reservation triggered — released on payment fail or timeout |
| Payment confirmed  | Cart cleared from Zustand + localStorage + Supabase, inventory decremented  |
| Gift card applied  | Gift card balance validated via API → partial deduction → remaining tracked |

## 8. Database Schema (PostgreSQL — Supabase)

### 8.1 Schema Overview

All tables: id (UUID PK), created\_at, updated\_at auto-managed by Supabase  
Row-Level Security (RLS) enabled on every table  
GST fields added per Indian law — HSN code + CGST/SGST/IGST amounts  
Idempotency keys stored on orders — prevents double-processing of webhooks

### 8.2 Users & Addresses

| Table     | Column                                                           | Type                 | Description                                  |
|-----------|------------------------------------------------------------------|----------------------|----------------------------------------------|
| users     | id                                                               | UUID PK              | Supabase Auth user ID                        |
| users     | email                                                            | VARCHAR<br>UNIQUE    | Primary login identity                       |
| users     | role                                                             | ENUM                 | customer editor manager admin super_admin    |
| users     | phone                                                            | VARCHAR(20)          | For OTP login                                |
| users     | loyalty_points                                                   | INTEGER<br>DEFAULT 0 | Current loyalty point balance                |
| users     | loyalty_tier                                                     | ENUM                 | bronze silver gold platinum                  |
| users     | referral_code                                                    | VARCHAR<br>UNIQUE    | Auto-generated unique referral code per user |
| users     | birthday                                                         | DATE                 | For birthday email campaign (optional)       |
| addresses | user_id                                                          | UUID FK              | References users.id                          |
| addresses | full_name,<br>phone, line1,<br>line2, city,<br>state,<br>pincode | Various              | Standard Indian address fields               |
| addresses | is_default                                                       | BOOLEAN              | Default delivery address                     |

### 8.3 Products, Variants & GST Fields

| Table    | Column                  | Type    | Description                                         |
|----------|-------------------------|---------|-----------------------------------------------------|
| products | name, slug,<br>base_sku | VARCHAR | Display name, URL slug, base SKU (e.g. SAM-CAN-001) |

|          |                                   |                |                                                           |
|----------|-----------------------------------|----------------|-----------------------------------------------------------|
| products | hsn_code                          | VARCHAR(10)    | ⚠ GST required — e.g. '3406' for candles                  |
| products | gst_rate                          | INTEGER        | ⚠ GST rate — 12 or 18 (percent). Candles = 12%            |
| products | price, sale_price                 | DECIMAL(10,2)  | Prices include GST (MRP model)                            |
| products | category_id, collection_id        | UUID FK        | Category and collection assignment                        |
| products | status                            | ENUM           | active draft archived out_of_stock                        |
| products | is_featured                       | BOOLEAN        | Appears on homepage featured section                      |
| products | weight_grams                      | INTEGER        | For Shiprocket shipping rate calculation                  |
| products | seo_title, seo_description        | VARCHAR        | Max 70 and 160 chars respectively                         |
| variants | sku                               | VARCHAR UNIQUE | e.g. SAM-CAN-001-180G — DB unique constraint              |
| variants | price, stock, low_stock_threshold | DECIMAL/INT    | Variant price, current stock, alert threshold (default 5) |
| variants | vessel_type, size_label           | ENUM / VARCHAR | glass ceramic terracotta   100g, 140g, 180g               |
| variants | barcode                           | VARCHAR(100)   | EAN-13 for label printing                                 |

## 8.4 Orders — GST & Idempotency Fields

| Table  | Column                                                   | Type           | Description                                                       |
|--------|----------------------------------------------------------|----------------|-------------------------------------------------------------------|
| orders | order_number                                             | VARCHAR UNIQUE | e.g. SAM-2025-0001                                                |
| orders | status                                                   | ENUM           | pending confirmed processing packed shipped delivered             |
| orders | payment_status                                           | ENUM           | pending paid failed refunded                                      |
| orders | idempotency_key                                          | VARCHAR UNIQUE | ⚠ Razorpay payment_id stored here — prevents duplicate processing |
| orders | subtotal, discount_amount, shipping_amount, total_amount | DECIMAL        | Financial breakdown                                               |
| orders | cgst_amount                                              | DECIMAL(10,2)  | ⚠ GST required — CGST portion (intra-state)                       |



|        |                                            |                   |                                                          |
|--------|--------------------------------------------|-------------------|----------------------------------------------------------|
| orders | sgst_amount                                | DECIMAL(10,2)     | ⚠ GST required — SGST portion (intra-state)              |
| orders | igst_amount                                | DECIMAL(10,2)     | ⚠ GST required — IGST (inter-state orders)               |
| orders | gift_note, gift_recipient, gift_occasion   | TEXT/VARCHAR      | Gift message data from cart                              |
| orders | razorpay_order_id, razorpay_payment_id     | VARCHAR           | Razorpay references                                      |
| orders | shiprocket_order_id, awb_number            | VARCHAR           | Shiprocket references                                    |
| orders | ndr_status                                 | ENUM nullable     | null delivery_failed re_attempt rto — for NDR management |
| orders | loyalty_points_earned, loyalty_points_used | INTEGER           | Points awarded and redeemed on this order                |
| orders | gift_card_id, gift_card_amount_used        | UUID FK / DECIMAL | Gift card redemption on this order                       |

## 8.5 Supporting Tables — New in v3.0

| Table                | Key Columns & Purpose                                                                                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| audit_logs           | id, user_id, action (e.g. 'price_changed'), entity_type, entity_id, before_data JSONB, ip_address, user_agent, created_at — full admin action trail                                           |
| gift_cards           | id, code VARCHAR UNIQUE, balance DECIMAL, original_amount, purchaser_user_id, expiry_date, is_active, used_amount, created_at                                                                 |
| loyalty_transactions | id, user_id, order_id FK nullable, type (earn redeem expire bonus), points_change, points_balance, created_at                                                                                 |
| referral_codes       | id, code UNIQUE (= users.referral_code), owner_user_id, referral_count, total_discount, created_at                                                                                            |
| referral_uses        | id, referral_code_id FK, referred_user_id, order_id FK, discount_amount, created_at, is_successful referral                                                                                   |
| instagram_gallery    | id, image_url (Cloudinary), caption TEXT, link_url, is_featured, sort_order, created_at, replaces live embed                                                                                  |
| homepage_banners     | id, slot_key (hero announcement brand_story campaign_left campaign_right testimonial newsletter), heading, subtext, image_url, cta_text, cta_url, bg_color, is_active, updated_at, updated_by |
| reviews              | id, product_id, user_id, rating(1-5), title, comment, images TEXT[], video_url, is_verified, is_approved, helpful_count, admin_reply, replied_at                                              |
| wholesale_customers  | id, company_name, contact_name, email, phone, gstin, moq INTEGER, custom_price, status (pending approved suspended), approved_by, approved_at                                                 |
| performance_audits   | id, audit_date, lighthouse_score, lcp_ms, cls_score, fcp_ms, broken_links_count, missing_images_count, low_stock_count, notes TEXT, run_by                                                    |

|            |                                                                                                                                                      |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| coupons    | id, code, type (percent fixed), value, min_order, max_uses, used_count, user_id nullable, auto_apply BOOLEAN                                         |
| blogs      | id, title, slug, content, cover_image_url, author_id, status (draft scheduled published), seo_description, tags TEXT[], categories TEXT[]            |
| cart       | id, user_id nullable, session_id (guests), product_id, variant_id, quantity, added_at, last_activity_at used by pg_cron for abandoned cart detection |
| newsletter | id, email, source, is_active, subscribed_at, unsubscribed_at, tags TEXT[] (for segments)                                                             |

## 9. Product & SKU Management

### 9.1 Product Categories & GST HSN Codes

| Category    | SKU Prefix | HSN Code | GST Rate                                     |
|-------------|------------|----------|----------------------------------------------|
| Candles     | SAM-CAN    | 3406     | 12% — confirmed for scented candles in India |
| Wax Melts   | SAM-WMT    | 3406     | 12%                                          |
| Room Sprays | SAM-RSP    | 3307     | 18% — room fresheners attract higher GST     |
| Bundles     | SAM-BND    | 3406     | 12% (majority component rule)                |
| Gift Sets   | SAM-GFT    | 3406     | 12%                                          |

⚠ IMPORTANT: HSN codes and GST rates above are based on standard classification. Samorah should confirm these with their Chartered Accountant before going live. Wrong HSN = invalid tax invoice.

### 9.2 SKU Format & Examples

| Category               | SKU Format            | Examples                                  |
|------------------------|-----------------------|-------------------------------------------|
| Candle 100g Glass      | SAM-CAN-[NUM]-100G-GL | SAM-CAN-001-100G-GL   SAM-CAN-002-100G-GL |
| Candle 140g Ceramic    | SAM-CAN-[NUM]-140G-CE | SAM-CAN-001-140G-CE                       |
| Candle 180g Terracotta | SAM-CAN-[NUM]-180G-TE | SAM-CAN-001-180G-TE                       |
| Wax Melt 50g           | SAM-WMT-[NUM]-50G     | SAM-WMT-001-50G                           |
| Room Spray 200ml       | SAM-RSP-[NUM]-200ML   | SAM-RSP-001-200ML                         |
| Bundle                 | SAM-BND-[NUM]         | SAM-BND-001                               |

### 9.3 Product Attributes for Filtering

| Attribute        | Values                                                       |
|------------------|--------------------------------------------------------------|
| Fragrance Family | Floral · Woody · Fresh · Dessert/Gourmand · Spicy · Oriental |

|                 |                                                            |
|-----------------|------------------------------------------------------------|
| Mood            | Calm · Warm · Festive · Fresh · Romantic · Cozy            |
| Vessel Material | Glass · Ceramic · Terracotta                               |
| Burn Time       | 35hr · 45hr · 55hr · 65hr                                  |
| Tags (badges)   | Bestseller · New Arrival · Limited · Seasonal · Staff Pick |
| HSN Code        | Per category — required for invoice                        |

## 10. Inventory Management

### 10.1 Core Features

| Feature           | Specification                                                                                            |
|-------------------|----------------------------------------------------------------------------------------------------------|
| Real-time Stock   | Tracked per variant SKU. Decrementated atomically on Razorpay payment.captured webhook                   |
| Stock Reservation | 15-minute hold on checkout initiation — Supabase pg_cron releases expired reservations                   |
| Low Stock Alert   | Admin email (Resend) when stock hits low_stock_threshold (default 5). Also shown in dashboard widget.    |
| Backorder         | Per-product allow_backorder flag — shows 'Ships in 7–10 days' at checkout if enabled                     |
| Bundle Stock      | Bundle availability = minimum stock of all 3 components. Alert fires on any component hitting threshold. |
| CSV Bulk Update   | Upload CSV (variant_sku + new_stock) → validation report → preview → confirm (atomic transaction)        |
| Return Restock    | Admin confirms return → stock restored → inventory_log entry (change_type: return)                       |
| Audit Trail       | inventory_logs: date   variant_id   change_type   qty_change   qty_after   note   admin_id               |

### 10.2 Inventory Admin Panel Screen

| Widget              | Content                                                                                                        |
|---------------------|----------------------------------------------------------------------------------------------------------------|
| Summary Cards       | Total active SKUs   In-stock   Low-stock   Out-of-stock                                                        |
| Low Stock Table     | SKU   Product   Variant   Current Stock   Threshold   Last Restock   Restock Now button                        |
| Stock Movements Log | Filter by: Date   SKU   Change Type   Admin. Columns: Date   SKU   Change Type   Qty Change   Qty After   Note |
| Bulk Restock Tool   | Upload CSV → validation errors highlighted → preview all changes → one-click confirm                           |
| Export Inventory    | CSV: SKU   Name   Variant   Price   Stock   Threshold   HSN   GST Rate                                         |

# 11. Admin Panel — Full Specification

## 11.1 Overview

URL: samorah.in/admin — protected by Next.js Middleware (role: editor+)  
Stack: Next.js App Router + Shadcn UI + TanStack Table + Recharts  
Layout: Left sidebar + topbar + main area — responsive, mobile-friendly

## 11.2 Dashboard

| Widget          | Content                                                            |
|-----------------|--------------------------------------------------------------------|
| Revenue Card    | Today's revenue   vs yesterday %   vs last week — green/red delta  |
| Orders Card     | Total orders today   Pending   Needs packing                       |
| Inventory Alert | Low stock count   Out-of-stock count   'Requires attention' link   |
| NDR Alert       | ⚠ New: Orders with delivery_failed status needing admin action     |
| Revenue Chart   | 30-day line chart — Recharts, hover for daily values               |
| Recent Orders   | Last 10 orders: Order#   Customer   Amount   Status   Quick update |
| Low Stock Table | Inline restock shortcut table                                      |
| Quick Actions   | Add Product   Create Coupon   View NDRs   Send Newsletter          |

## 11.3 Product Management (6-Step Form)

| Step | Name            | Fields                                                                                         |
|------|-----------------|------------------------------------------------------------------------------------------------|
| 1    | Basic Info      | Name, slug (auto), category, collection, price, sale_price, HSN code, GST rate, weight, status |
| 2    | Variants        | Add/remove rows: variant_name, SKU, price, stock, vessel_type, size_label, barcode, is_active  |
| 3    | Images          | Cloudinary drag-drop upload, reorder, set primary, add alt text (required)                     |
| 4    | Fragrance Notes | Top / Heart / Base — tag-style input, multiple notes per layer                                 |
| 5    | SEO             | seo_title (70 char count), seo_description (160 char count), SERP snippet preview              |
| 6    | Preview         | Live render of product card and product page hero                                              |

## 11.4 Order Management + Tax Invoice

| Feature         | Specification                                                                                                                                                   |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Order List      | Order#   Customer   Date   Items   Amount   GST   Payment Status   Order Status   Actions                                                                       |
| Order Detail    | Items + images, delivery address, payment ref, Razorpay ID, Shiprocket AWB, order timeline, gift note                                                           |
| Tax Invoice PDF | ⚠ Must be a legal 'Tax Invoice' — includes: seller GSTIN, buyer name/address, HSN per line item, taxable value, CGST/SGST or IGST, invoice number, invoice date |
| Status Update   | Dropdown change → automated Resend email per transition                                                                                                         |
| Shiprocket      | Create shipment → assign courier → AWB → print label → schedule pickup                                                                                          |
| Refund          | Razorpay refund (full/partial) → restock if returned → refund email                                                                                             |
| Internal Notes  | Admin-only timestamped notes — not visible to customer                                                                                                          |
| NDR Tab         | ⚠ New: Dedicated NDR view — delivery_failed orders   'Instruct Re-attempt' button   'Accept RTO' button                                                         |

## 11.5 Coupon Management

| Feature          | Specification                                                                                                                           |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Create Coupon    | Code   Type (% or ₹)   Value   Min order   Max uses   User-specific   Product restriction   Expiry   First-order-only   Auto-apply flag |
| Usage Analytics  | Redemptions   Total discount given   Revenue from coupon users   Who used it                                                            |
| Welcome Coupon   | Auto-generated first-order coupon in welcome email — % configurable in Settings                                                         |
| Referral Coupons | Auto-generated when referral link leads to first purchase — ₹100 off by default                                                         |

## 11.6 Gift Card Management

| Feature            | Specification                                                                                    |
|--------------------|--------------------------------------------------------------------------------------------------|
| Generate Gift Card | Admin creates: amount, expiry, recipient email (sends branded delivery email)                    |
| Customer Purchase  | Customer buys gift card on /gift-cards — Razorpay payment → code generated → delivery email sent |

|                |                                                                                |
|----------------|--------------------------------------------------------------------------------|
| Redemption     | Applied at cart — code validated → balance checked → partial deduction tracked |
| Gift Card List | Code   Amount   Balance   Purchaser   Recipient   Expiry   Status              |

## 11.7 Loyalty & Referral Management

| Feature              | Specification                                                                              |
|----------------------|--------------------------------------------------------------------------------------------|
| Points Config        | Earn rate: X points per ₹100 spent   Redeem rate: Y points = ₹1 — configurable in settings |
| Tier Thresholds      | Bronze: 0pts   Silver: 500pts   Gold: 1500pts   Platinum: 5000pts — configurable           |
| Customer Points View | Per customer: current balance, tier, transaction history, lifetime earned                  |
| Referral Config      | Referrer reward: ₹100 coupon   Referred reward: ₹100 off first order — configurable        |
| Manual Adjustment    | Super Admin can add/deduct points manually with reason note                                |

## 11.8 Review Management

| Feature          | Specification                                                                               |
|------------------|---------------------------------------------------------------------------------------------|
| Review List      | Product   Rating   Customer   Date   Verified Purchase   Status (pending/approved/rejected) |
| Moderation Queue | New reviews land in pending state — admin approves/rejects before visible on site           |
| Admin Reply      | Reply field per review — displayed publicly below customer review                           |
| Image/Video      | Customers can upload images (Cloudinary) and video URL with reviews                         |
| Helpful Votes    | Customers mark reviews as helpful — sort by most helpful in product page                    |
| Export Reviews   | CSV export for external analysis or Klaviyo segmentation                                    |

## 11.9 Blog / Journal Management

| Feature     | Specification                                                                             |
|-------------|-------------------------------------------------------------------------------------------|
| Rich Editor | MDX / Sanity rich text — image upload, embed product cards, headings, tables, code blocks |



|                  |                                                                                          |
|------------------|------------------------------------------------------------------------------------------|
| Draft / Schedule | Save as draft   Schedule publish date + time   Publish immediately   Unpublish           |
| SEO Fields       | SEO title   Meta desc   Custom slug   OG image   Tags   Categories                       |
| Campaign Tags    | Tag posts as: Diwali   Christmas   Valentine   Mother's Day — for email campaign linking |

## 11.10 SEO, Media Library & Analytics

| Section       | Key Features                                                                                                                                      |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| SEO Manager   | Default meta template   Sitemap download   Robots.txt editor   Redirect manager (301/302)   SEO audit (missing meta, no-alt images, broken links) |
| Media Library | Cloudinary drag-drop   Search by filename   Alt text editor   Usage tracking (prevents deleting in-use images)                                    |
| Analytics     | Sales report   Product performance   Customer insights   Coupon performance   Inventory report   GA4 embed                                        |

## 11.11 Settings — Super Admin Only

| Setting Group      | Options                                                                                     |
|--------------------|---------------------------------------------------------------------------------------------|
| Store              | Name   Logo   Currency INR   Timezone IST   Email sender   GST number   Registered address  |
| Tax / GST          | Store GSTIN   Default state (for CGST/SGST vs IGST logic)   Tax-inclusive pricing toggle    |
| Payments           | Razorpay Key ID (masked)   Payment methods on/off   COD enable   Test mode toggle           |
| Shipping           | Shiprocket credentials   Free shipping threshold   Pickup address   NDR auto-retry settings |
| Email Templates    | Edit all 15 email templates from admin — no code changes required                           |
| Loyalty / Referral | Points earn rate   Redeem rate   Tier thresholds   Referral reward amounts                  |
| Backup             | Manual trigger: DB export   Products CSV   Orders CSV   Last backup timestamp               |
| Audit Logs         | Full log: timestamp   user   action   entity   before JSONB   after JSONB   IP              |



## 12. Payment Integration — Razorpay

### 12.1 Supported Payment Methods

| Method      | Details                                         | India Coverage                           |
|-------------|-------------------------------------------------|------------------------------------------|
| UPI         | GPay, PhonePe, Paytm, BHIM, UPI QR              | #1 India payment — 70%+ digital payments |
| RuPay Cards | Debit + credit, RuPay on UPI                    | Government-promoted, very common         |
| Cards       | Visa, Mastercard, AmEx — debit + credit         | Domestic + international                 |
| Net Banking | 50+ banks: SBI, HDFC, ICICI, Axis               | Preferred for high-value orders          |
| Wallets     | Paytm, PhonePe, Amazon Pay                      | Common among younger buyers              |
| EMI         | No-cost EMI — 3/6/9/12 months on cards          | Increases AOV on bundles                 |
| COD         | Cash on delivery — pincode-based via Shiprocket | Trust-building for first-time buyers     |

### 12.2 Payment Flow

1. Customer clicks 'Place Order' → Next.js API: POST /api/razorpay/create-order
2. Razorpay order created with amount (paise), currency INR, receipt ID — stored in DB
3. Razorpay checkout overlay opens — pre-filled name, email, phone
4. Customer pays — Razorpay returns payment\_id, order\_id, signature to frontend
5. Frontend sends all 3 to POST /api/razorpay/verify
6. API verifies HMAC SHA256(order\_id + '|' + payment\_id) using key\_secret
7. Signature valid → order updated → inventory decremented → Shiprocket triggered → email sent

### 12.3 Critical: Webhook Idempotency

⚠ CRITICAL: Razorpay and Shiprocket will sometimes send the SAME webhook event TWICE due to network retries.

The Problem: If your handler decrements inventory on every payment.captured event, a duplicate webhook will deduct stock TWICE — causing phantom inventory loss.

The Fix — implement in /api/razorpay/webhook:

1. On receiving payment.captured, extract razorpay\_payment\_id
2. Check: SELECT id FROM orders WHERE idempotency\_key = razorpay\_payment\_id
3. If row exists → order already processed → return 200 OK immediately (do nothing else)
4. If no row → this is the first processing → proceed with inventory, shipping, email
5. Set orders.idempotency\_key = razorpay\_payment\_id BEFORE triggering downstream actions

Same pattern applies to Shiprocket webhook status updates.

## 12.4 Webhook Events

| Event            | System Action (after idempotency check)                                                                     |
|------------------|-------------------------------------------------------------------------------------------------------------|
| payment.captured | Order → paid, inventory decremented, GST calculated, Shiprocket order created, confirmation + invoice email |
| payment.failed   | Order → failed, reserved stock released, failure email, customer shown retry option                         |
| refund.created   | Log refund, order → refunded, refund initiated email                                                        |
| refund.processed | Refund status updated, inventory restocked if return, refund complete email                                 |

## 13. Shipping Integration — Shiprocket + NDR Management

### 13.1 Core Integration Flow

8. Payment confirmed → POST /v1/external/orders/create to Shiprocket
9. shiprocket\_order\_id stored → Admin assigns courier (or auto-assign)
10. AWB generated → orders.awb\_number stored → 'Shipped' email triggered
11. Pickup scheduled → Shiprocket webhook sends status updates
12. Real-time tracking: Supabase updated on each webhook → /account/orders/[id] shows timeline

### 13.2 Critical: NDR Management (Non-Delivery Reports)

⚠ HIGH RISK IN INDIA: A significant percentage of COD orders result in RTO (Return to Origin)

because the customer is unavailable, rejects the package, or the address is incorrect.

Without NDR management, these failed deliveries go unnoticed and result in:

- Lost revenue (product shipped but not delivered, not returned quickly)
- Increased RTO rate (hurts Shiprocket service rating)
- Customer dissatisfaction (no communication)

Required Admin Panel Feature — NDR Tab in Orders section:

- Dedicated view: all orders where ndr\_status = 'delivery\_failed'
- Columns: Order# | Customer | AWB | Courier | Failure Reason | Failure Date | Action
- 'Instruct Re-attempt' button → calls Shiprocket NDR API to schedule re-delivery
- 'Accept RTO' button → marks order for return, triggers refund flow
- Shiprocket NDR webhook updates orders.ndr\_status automatically

### 13.3 Shipping Features

| Feature         | Description                                                         |
|-----------------|---------------------------------------------------------------------|
| Rate Calculator | Real-time rate at checkout — pincode + package weight from variants |
| Multi-courier   | Auto-select best courier: Blue Dart, Delhivery, DTDC, Ecom Express  |
| NDR Management  | Admin panel NDR tab — re-attempt or RTO per failed delivery         |

|                    |                                                                        |
|--------------------|------------------------------------------------------------------------|
| Tracking           | Branded tracking page at samorah.in/track/[awb] or embedded in account |
| Cancellation       | Before pickup: cancel → restock → Razorpay refund → email              |
| COD Reconciliation | COD remittance tracking — Shiprocket reports deposit date              |

## 13.4 Shipping Rules

| Rule              | Config                                                                     |
|-------------------|----------------------------------------------------------------------------|
| Free Shipping     | Above ₹1,000 — configurable in Admin → Settings → Shipping                 |
| Standard Delivery | 5–7 business days — auto-selected best rate                                |
| Express Delivery  | 2–3 days — Blue Dart / Delhivery express — premium rate passed to customer |
| COD Availability  | Pincode-based from Shiprocket — shown/hidden dynamically at checkout       |

## 14. Email System & Marketing Automation

### 14.1 Technology

Primary: Resend + React Email — transactional emails (free: 3,000/month)

Marketing sequences: Resend Broadcasts — or Brevo free tier (300 emails/day)

Scheduled triggers: Supabase pg\_cron sweeps cart table hourly for abandoned carts

All templates editable from Admin → Settings → Email Templates (no code changes)

### 14.2 Transactional Email Templates (15)

| Email                     | Trigger                   | Key Content                                                                                       |
|---------------------------|---------------------------|---------------------------------------------------------------------------------------------------|
| Welcome                   | Registration              | Brand welcome + first-order coupon + explore collections CTA                                      |
| Email Verification<br>OTP | Register / OTP<br>request | 6-digit OTP, 10-min expiry, resend link                                                           |
| Password Reset            | Forgot password           | Secure link — 30-min expiry, branded<br>button                                                    |
| Order Confirmed           | payment.captured          | Order#   items with images   GST<br>breakdown   delivery estimate   Tax Invoice<br>PDF attachment |
| Order Packed              | Status → packed           | Packed and ready, expected dispatch date                                                          |
| Order Shipped             | AWB generated             | AWB#   Courier   Live tracking link  <br>Estimated delivery                                       |
| Delivered                 | Status → delivered        | Thank you   Write a review CTA   Candle<br>care reminder                                          |
| Review Request            | 3 days after delivered    | Product image   One-click star rating  <br>Review form link                                       |
| Refund Initiated          | Refund created            | Amount   Timeline 5–7 days   Order<br>reference                                                   |
| Gift Card Delivery        | Gift card purchased       | Gift card code   Balance   Expiry   How to<br>redeem                                              |
| Loyalty Points<br>Earned  | Order delivered           | Points earned   New balance   Tier status  <br>Redeem CTA                                         |
| Tier Upgrade              | Tier threshold<br>crossed | New tier name (Gold!)   Benefits   Next tier<br>milestone                                         |
| Newsletter Welcome        | Newsletter signup         | Brand story   Best sellers   Unsubscribe link                                                     |

|                  |                        |                                                                |
|------------------|------------------------|----------------------------------------------------------------|
| Low Stock Alert  | Stock hits threshold   | Admin email: SKU   Product   Variant   Current count           |
| NDR Notification | Shiprocket NDR webhook | To customer: delivery failed notice   contact / re-attempt CTA |

## 14.3 Marketing Automation Sequences

### Abandoned Cart Sequence (pg\_cron sweeps hourly)

| Email # | Trigger              | Content                                                                                      |
|---------|----------------------|----------------------------------------------------------------------------------------------|
| Email 1 | 1 hour after abandon | Subject: 'Your candles are waiting' — cart items with images, soft reminder, no discount     |
| Email 2 | 24 hours after       | Subject: 'Still thinking about us?' — cart items, fragrance story excerpt, no discount       |
| Email 3 | 72 hours after       | Subject: 'A small gift for you' — cart items + 10% off coupon auto-generated, expires in 48h |

Mechanism: Supabase pg\_cron runs every hour: SELECT carts WHERE last\_activity\_at < NOW()-INTERVAL '1 hour' AND NOT EXISTS (order placed). Triggers Resend API for each matching cart.

### Welcome Sequence (new registrations)

| Email # | Delay     | Content                                                   |
|---------|-----------|-----------------------------------------------------------|
| Email 1 | Immediate | Welcome + first-order coupon (WELCOME10 or WELCOME20)     |
| Email 2 | Day 3     | Brand story — 'Why we make candles' — editorial narrative |
| Email 3 | Day 7     | 'Meet the Chapters' — introduce Vol. I–IV with shop links |

### Post-Purchase Sequence

| Email # | Delay                | Content                                                                                |
|---------|----------------------|----------------------------------------------------------------------------------------|
| Email 1 | Order confirmed      | Confirmation + Tax Invoice PDF                                                         |
| Email 2 | 3 days post-deliver  | Review request — product image, star rating, review link                               |
| Email 3 | 30 days post-deliver | Replenishment nudge — 'Your candle may be running low' + reorder CTA + recommendations |



## Festival Campaign Schedule

| Campaign          | Timing                | Content Focus                                                       |
|-------------------|-----------------------|---------------------------------------------------------------------|
| Diwali Collection | 3 weeks before Diwali | Festival-themed candles, gift sets, bundle offers, free shipping    |
| Christmas         | Dec 1–25              | Festive chapter, gift cards, corporate gifting CTA                  |
| Valentine's Day   | Feb 1–14              | Romantic fragrance family, bundle builder feature, gift wrap option |
| Mother's Day      | 1 week before         | Gift curation, gift card feature, personalized message option       |
| Volume Release    | New collection launch | Pre-launch teaser → launch day announcement → restock alert         |
| Birthday Campaign | Day of birthday       | Surprise gift: bonus loyalty points or exclusive coupon             |

## 15. Loyalty System & Referral Program

### 15.1 Loyalty Points System

| Parameter        | Default Value (configurable in Admin Settings)          |
|------------------|---------------------------------------------------------|
| Earn Rate        | 10 points per ₹100 spent (configurable)                 |
| Redeem Rate      | 100 points = ₹10 off (configurable)                     |
| Points on Signup | 50 bonus points for creating account                    |
| Points Expiry    | 1 year from earn date — expires if not used             |
| Review Bonus     | 25 points for approved product review                   |
| Birthday Bonus   | 100 points auto-credited on birthday (if date provided) |

### 15.2 Loyalty Tiers

| Tier     | Points Required | Benefits                                                          | Badge Color        |
|----------|-----------------|-------------------------------------------------------------------|--------------------|
| Bronze   | 0–499           | Standard earn rate                                                | Warm tan (#9C826B) |
| Silver   | 500–1499        | 1.2x earn rate + early access to new collections                  | Silver (#A8A8A8)   |
| Gold     | 1500–4999       | 1.5x earn rate + free standard shipping on all orders             | Gold (#C8A951)     |
| Platinum | 5000+           | 2x earn rate + free express shipping + exclusive limited releases | Platinum (#818181) |

### 15.3 Referral Program

| Component       | Specification                                                                       |
|-----------------|-------------------------------------------------------------------------------------|
| Referral Code   | Every user gets a unique referral_code (e.g. SARA100) — shown in /account dashboard |
| Share Mechanism | Copy link button: samorah.in?ref=SARA100   WhatsApp share   Email share             |
| Referrer Reward | ₹100 coupon auto-generated when referred user completes first order                 |

|                 |                                                                             |
|-----------------|-----------------------------------------------------------------------------|
| Referred Reward | ₹100 off first order — applied automatically via ref= URL parameter         |
| Tracking        | referral_uses table: tracks who referred whom, which order, discount amount |
| Admin View      | Referral leaderboard: top referrers by count and revenue generated          |

## 15.4 Customer Account — Loyalty Dashboard (/account/loyalty)

- Current points balance (large, prominent display)
- Current tier badge + progress bar to next tier
- Points transaction history: date, type (earn/redeem/bonus), amount, order link
- Redemption widget: enter points to redeem → converts to ₹ discount → apply to cart
- Your referral link with copy button and WhatsApp share
- Referral history: friends referred, date, reward earned

## 16. Gift Cards & Gift Messages

### 16.1 Gift Card System

| Feature          | Specification                                                                                                                                   |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Purchase         | Customer goes to /gift-cards → selects amount (₹500 / ₹1000 / ₹2000 / custom) → Razorpay payment → code generated → delivery email to recipient |
| Code Format      | SAM-GC-[8 random chars] — e.g. SAM-GC-X7K2M9PT                                                                                                  |
| Balance Tracking | Partial redemption supported — remaining balance preserved until expiry                                                                         |
| Expiry           | 12 months from purchase date — shown in delivery email                                                                                          |
| Application      | Applied at cart — code field → validated → balance deducted → remaining tracked                                                                 |
| Admin            | Admin can manually generate gift cards (e.g. for customer service resolution)                                                                   |
| Recipient Email  | Branded email: gift card code, amount, expiry, personal message from sender, redeem CTA                                                         |

### 16.2 Gift Messages (at Checkout)

| Field          | Specification                                                                                                 |
|----------------|---------------------------------------------------------------------------------------------------------------|
| Gift Note      | Free text, max 200 chars — printed on packing slip, shown in order detail                                     |
| Recipient Name | Optional — 'To: [name]' printed on packing slip                                                               |
| Occasion       | Dropdown: Birthday   Anniversary   Diwali   Christmas   Mother's Day   Just Because   Other                   |
| Gift Wrap      | Phase 2 — checkbox for branded Samorah box (+₹99)                                                             |
| Packing Slip   | When gift note present: packing slip shows 'Gift Order' header, excludes pricing, shows gift note prominently |

## 17. Homepage Banner Manager & Instagram Gallery Admin

### 17.1 Why a Banner Manager

Without a Banner Manager, every content change on the homepage requires a developer. With this system, Editors can update hero headlines, campaign blocks, testimonials, and seasonal content in 2 minutes — without touching code or calling a developer. This is role: editor+ in the admin panel — no Super Admin required.

### 17.2 Homepage Banner Slots

| Slot Key            | Role Required | Editable Fields                                           |
|---------------------|---------------|-----------------------------------------------------------|
| announcement        | Editor+       | text, link_url, bg_color, is_active                       |
| hero                | Editor+       | heading, subtext, cta_text, cta_url, image_url, is_active |
| brand_story         | Editor+       | heading, copy_text, cta_text, cta_url, image_url          |
| campaign_left       | Editor+       | label, title, subtext, cta_text, cta_url, image_url       |
| campaign_right      | Editor+       | label, title, subtext, cta_text, cta_url, image_url       |
| testimonial         | Editor+       | quote_text, attribution                                   |
| newsletter_headline | Editor+       | headline, subtext                                         |
| footer_quote        | Editor+       | quote_text                                                |

### 17.3 Banner Manager Admin Screen

- URL: /admin/banners
- Left panel: List of all slots with current preview and last updated timestamp
- Right panel: Editing form for selected slot — image upload via Cloudinary
- Live preview button: see changes before publishing
- Publish button: updates homepage\_banners table, ISR revalidation triggered for homepage
- Change history: who updated what and when — via audit\_logs

### 17.4 Instagram Gallery Admin

WHY: Embedding Instagram's live feed requires a third-party API key (costs ₹2,000+/month) and breaks regularly when Instagram changes their API.

SOLUTION: Admin-managed gallery table. Editors manually upload curated lifestyle images to the `instagram_gallery` table. These are displayed in the homepage atmosphere grid. Images are pre-curated, always available, and require no external API.

| Feature       | Specification                                                                          |
|---------------|----------------------------------------------------------------------------------------|
| Upload        | Drag-drop image upload to Cloudinary → auto-resized to 1:1 grid                        |
| Caption       | Optional caption text — displayed on hover                                             |
| Link URL      | Optional link when image clicked — can point to product, collection, or Instagram post |
| Sort Order    | Drag to reorder — determines grid layout                                               |
| Featured Flag | Mark top 9 images as featured — only featured images shown on homepage                 |
| Grid Style    | Alternating sizes on homepage — 3 large + 6 small in magazine layout                   |
| Admin URL     | /admin/instagram-gallery                                                               |

## 18. Event Tracking Architecture (GA4 + Meta Pixel)

### 18.1 GA4 Custom Events

| GA4 Event Name          | Trigger                 | Parameters                                                          |
|-------------------------|-------------------------|---------------------------------------------------------------------|
| view_item               | Product page load       | item_id (SKU), item_name, item_category, price, currency (INR)      |
| add_to_cart             | Add to Cart click       | item_id, item_name, item_variant (size), price, quantity            |
| remove_from_cart        | Remove from cart        | item_id, item_name, price, quantity                                 |
| view_cart               | Cart page open          | cart_value, item_count, has_coupon                                  |
| begin_checkout          | Proceed to checkout     | cart_value, coupon, item_count                                      |
| add_shipping_info       | Delivery step           | shipping_tier (standard/express), shipping_cost                     |
| add_payment_info        | Payment step            | payment_type (upi/card/cod/wallet)                                  |
| purchase                | Order confirmed         | transaction_id, value, tax (GST total), shipping, currency, items[] |
| search                  | Search query submit     | search_term, result_count                                           |
| view_item_list          | Collection / shop page  | item_list_name (collection name), items[]                           |
| select_item             | Click product from list | item_list_name, item_id, item_position                              |
| add_to_wishlist         | Wishlist heart click    | item_id, item_name, price                                           |
| review_submitted        | Review form submit      | item_id, rating                                                     |
| newsletter_signup       | Newsletter form submit  | source (homepage/product/checkout)                                  |
| bundle_started          | Bundle builder opened   | vessel_filter selected                                              |
| bundle_completed        | Bundle added to cart    | item_ids (3 candles), total_value, savings                          |
| coupon_applied          | Coupon code success     | coupon_code, discount_amount                                        |
| gift_card_applied       | Gift card redeemed      | gift_card_amount                                                    |
| loyalty_points_redeemed | Points applied at cart  | points_used, discount_value                                         |

|                      |                        |                               |
|----------------------|------------------------|-------------------------------|
| referral_link_shared | Share referral clicked | channel (whatsapp/copy/email) |
|----------------------|------------------------|-------------------------------|

## 18.2 Meta Pixel Events

| Meta Event           | Maps To                                                 |
|----------------------|---------------------------------------------------------|
| ViewContent          | view_item — for product retargeting ads                 |
| AddToCart            | add_to_cart — for cart abandonment retargeting          |
| InitiateCheckout     | begin_checkout — for checkout drop-off campaigns        |
| Purchase             | purchase — for ROAS calculation and lookalike audiences |
| CompleteRegistration | Account registration — for acquisition campaigns        |

## 18.3 Microsoft Clarity (Heatmaps)

- Session recordings to understand user behavior on product pages and checkout
- Click heatmaps — identify high-engagement vs ignored sections
- Scroll depth — see how far users read on collection pages and blog posts
- Rage click and dead click detection — identify UX friction points
- Free tool — no cost, no setup required beyond adding script tag



## 19. Backup Strategy & Disaster Recovery

### 19.1 Backup Schedule

| Frequency | What                         | Method                               | Retention                    |
|-----------|------------------------------|--------------------------------------|------------------------------|
| Daily     | Supabase database (auto)     | Supabase Cloud built-in              | 7 days (free), 30 days (Pro) |
| Daily     | Admin-triggered full DB dump | Admin → Settings → Backup → Trigger  | Stored in Supabase Storage   |
| Weekly    | Cloudinary media library     | Cloudinary backup download           | Keep last 4 weeks            |
| Weekly    | GitHub code push             | Auto via CI/CD (every deploy)        | Git history permanent        |
| Monthly   | Products CSV export          | Admin → Products → Export CSV        | Keep 6 months                |
| Monthly   | Customers CSV export         | Admin → Customers → Export CSV       | Keep 6 months                |
| Monthly   | Orders CSV export            | Admin → Orders → Export CSV          | Keep 12 months (GST records) |
| Monthly   | Reviews, Blogs, Coupons CSV  | Admin → respective sections → Export | Keep 6 months                |

### 19.2 Disaster Recovery

| Scenario                  | Recovery Procedure                                                                |
|---------------------------|-----------------------------------------------------------------------------------|
| Database corruption       | Supabase restores from last daily backup — RTO < 30 minutes                       |
| Accidental data deletion  | Restore from Supabase point-in-time recovery — select timestamp before deletion   |
| Vercel deployment failure | Vercel auto-rollback to last stable deployment — instant, no manual action needed |
| Cloudinary image loss     | Re-upload from local copies (Samorah must keep original images locally)           |
| Domain issue              | GoDaddy DNS records documented — re-point to Vercel in < 1 hour                   |
| Recovery Time Objective   | < 30 minutes for any failure scenario (database, deployment, domain)              |
| Recovery Point Objective  | < 24 hours — maximum data loss = 1 day of transactions                            |

### 19.3 Admin Backup Screen

- URL: /admin/settings → Backup section
- Buttons: 'Export Database' | 'Export Products CSV' | 'Export Orders CSV' | 'Export Customers CSV'
- Last backup timestamps displayed for each type
- Supabase auto-backup status indicator (green = active, red = check Supabase dashboard)
- Documentation link: step-by-step recovery instructions stored in admin docs

## 20. Audit Logs

### 20.1 What Gets Logged

| Category   | Action                        | Before / After Data                                 |
|------------|-------------------------------|-----------------------------------------------------|
| Products   | Price changed                 | Who   When   Old price   New price                  |
| Products   | Stock adjusted                | Who   When   Old stock   New stock   Reason         |
| Products   | Status changed (draft→active) | Who   When   Old status   New status                |
| Orders     | Status updated                | Who   When   Old status   New status                |
| Orders     | Refund initiated              | Who   When   Amount   Razorpay refund ID            |
| Coupons    | Created / paused / deleted    | Who   When   Full coupon config                     |
| Users      | Role changed                  | Who changed   Who was changed   Old role   New role |
| Users      | Account suspended             | Who   When   Reason                                 |
| Banners    | Homepage section updated      | Who   When   Slot   Old content   New content       |
| Settings   | Any settings change           | Who   When   Setting key   Old value   New value    |
| Gift Cards | Manually generated            | Who   When   Amount   Recipient                     |
| Loyalty    | Manual point adjustment       | Who adjusted   Customer   Points change   Reason    |
| Blog       | Published / unpublished       | Who   When   Post slug                              |

### 20.2 Audit Log Schema

Table: audit\_logs

| Column      | Type         | Notes                                                  |
|-------------|--------------|--------------------------------------------------------|
| id          | UUID PK      | Auto-generated                                         |
| user_id     | UUID FK      | Admin who performed the action                         |
| action      | VARCHAR(100) | e.g. 'price_changed', 'order_refunded', 'role_changed' |
| entity_type | VARCHAR(50)  | e.g. 'product', 'order', 'user', 'coupon'              |
| entity_id   | UUID         | ID of the affected record                              |

|             |             |                                                |
|-------------|-------------|------------------------------------------------|
| before_data | JSONB       | State before the change — full record snapshot |
| after_data  | JSONB       | State after the change — full record snapshot  |
| ip_address  | VARCHAR(45) | Admin's IP — for security monitoring           |
| user_agent  | TEXT        | Browser/OS — for security monitoring           |
| created_at  | TIMESTAMP   | Auto-set by Supabase                           |

## 20.3 Audit Log Admin Screen

- URL: /admin/audit-logs — Super Admin only
- Filters: Date range | User (who acted) | Action type | Entity type
- Expandable rows: click to see full before/after JSON diff highlighted
- Export to CSV for compliance records
- Cannot delete audit logs — immutable records

## 21. Monthly Performance Audit

### 21.1 Audit Checklist

| Category    | Metric                 | Target / Action                                                |
|-------------|------------------------|----------------------------------------------------------------|
| SEO         | Broken links           | 0 broken links — fix any 404s with 301 redirects in admin      |
| SEO         | Missing metadata       | 0 products without seo_title and seo_description               |
| SEO         | Missing alt text       | 0 product images without alt_text                              |
| SEO         | 404 pages              | Review GA4 Behavior → 404 report — redirect all                |
| Performance | Lighthouse Performance | Target: >90 on mobile + desktop                                |
| Performance | Core Web Vitals — LCP  | Target: <2.5 seconds                                           |
| Performance | Core Web Vitals — CLS  | Target: <0.1                                                   |
| Performance | Core Web Vitals — FCP  | Target: <1.8 seconds                                           |
| Inventory   | Low stock SKUs         | Review and restock before stockout                             |
| Inventory   | Dead inventory         | Products with 0 sales in 60 days — consider promotions         |
| Inventory   | Best sellers           | Ensure top 10 SKUs always in stock                             |
| Marketing   | Top pages              | GA4 — which collection/product pages drive most sessions       |
| Marketing   | Conversion rate        | Sessions → Add to Cart → Checkout → Purchase funnel            |
| Marketing   | Email open rates       | Resend dashboard — open rate target >35% for transactional     |
| Marketing   | Cart abandonment rate  | Target: <70% — check abandoned cart email sequence performance |
| Marketing   | NDR rate               | Target: <5% — high NDR hurts Shiprocket service rating         |

### 21.2 Audit Log in Admin

- URL: /admin/performance-audits
- Monthly audit results stored in performance\_audits table — historical tracking

- Lighthouse scores can be pasted manually or triggered via Vercel integration
- Broken link report generated by admin utility (crawls all sitemap URLs)

## 22. Review System — Enhanced

### 22.1 Customer-Facing Review Features

| Feature        | Specification                                                                                 |
|----------------|-----------------------------------------------------------------------------------------------|
| Write Review   | Available only after delivery confirmed (is_verified_purchase = true) — prevents fake reviews |
| Rating         | 1–5 star scale — required field                                                               |
| Review Title   | Optional short title, max 100 chars                                                           |
| Review Body    | Required, min 50 chars, max 1000 chars                                                        |
| Image Upload   | Up to 3 images — Cloudinary upload, shown in review card                                      |
| Video URL      | Optional YouTube / Reel link — embedded in review                                             |
| Verified Badge | Green 'Verified Purchase' badge if is_verified_purchase = true                                |
| Helpful Votes  | 'Was this helpful?' Yes/No — increments helpful_count per review                              |
| Sort Options   | Most helpful   Most recent   Highest rated   Lowest rated                                     |
| Admin Reply    | Displayed below customer review with 'Samorah Response' label                                 |

### 22.2 Review Flow & Moderation

13. Customer writes review → submitted → status: pending
14. Admin sees pending reviews in moderation queue (/admin/reviews)
15. Admin approves → status: approved → visible on product page immediately
16. Admin replies → reply stored in admin\_reply field → displayed publicly
17. Review request email sent 3 days after delivery — one-click star rating in email
18. Approved reviews used in structured data (JSON-LD) for SEO aggregate rating

## 23. SEO, Performance & Security

### 23.1 SEO Implementation

| Feature         | Implementation                                                                      |
|-----------------|-------------------------------------------------------------------------------------|
| Dynamic Meta    | Next.js 15 generateMetadata() — title, description, canonical, OG, Twitter per page |
| Structured Data | JSON-LD: Product (with aggregateRating), BreadcrumbList, Organization, BlogPosting  |
| GST Schema      | Product JSON-LD includes price excluding tax for Google Shopping compatibility      |
| Sitemap         | Auto-generated sitemap.xml — products, collections, blog posts, static pages        |
| Robots.txt      | Block: /admin, /cart, /checkout, /account. Allow: everything else                   |
| Target Score    | Lighthouse: Performance >90, SEO >95, Best Practices >95, Accessibility >90         |

### 23.2 Performance Techniques

| Technique    | Detail                                                                    |
|--------------|---------------------------------------------------------------------------|
| ISR          | Product pages re-generated every 60s — fast + fresh without SSR overhead  |
| Image        | next/image + Cloudinary CDN — WebP/AVIF auto-serve, blur placeholder      |
| Font         | next/font — self-hosted Google Fonts, preloaded, no FOUT                  |
| Lazy loading | Below-fold images, heavy components, third-party scripts                  |
| pg_cron jobs | Abandon cart sweeps run in DB — no server resources used on the Node side |

### 23.3 Security Layers

| Layer            | Implementation                                                                        |
|------------------|---------------------------------------------------------------------------------------|
| HTTPS            | Auto-managed Vercel SSL — all HTTP→HTTPS enforced                                     |
| Webhook Security | Razorpay: HMAC SHA256 verify   Shiprocket: shared secret header   + idempotency check |
| Auth Tokens      | Access token in memory only   Refresh in httpOnly Secure SameSite=Strict cookie       |



|                    |                                                                                                                                                                                                                                   |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CSRF               | SameSite=Strict cookies   CSRF token on all state-changing requests                                                                                                                                                               |
| XSS                | React auto-escapes JSX   Content-Security-Policy header   Never dangerouslySetInnerHTML                                                                                                                                           |
| SQL Injection      | Supabase parameterized queries only — zero raw SQL string concatenation                                                                                                                                                           |
| Rate Limiting      | Vercel Edge rate limiting on auth, checkout, contact form, coupon apply endpoints. <i>Strict IP-based rate limiting on the /api/auth/otp request endpoint (e.g., max 3 requests per 15 minutes) to prevent SMS bombing fraud"</i> |
| DPDP Compliance    | India Data Protection Act — consent on signup, data minimization, right to deletion                                                                                                                                               |
| GST Record Keeping | Orders and invoice data retained 7 years — required by Indian tax law                                                                                                                                                             |

## 24. Critical Technical Edge Cases & Fixes

This section documents 5 critical technical issues that WILL cause bugs or crashes if not addressed.

Reference this section when prompting Claude for each affected phase.

### 24.1 Next.js + Zustand Hydration Mismatch (Affects Phase 6)

Problem: Next.js renders on server first. localStorage doesn't exist on server.

Result: Zustand with persist(localStorage) throws Hydration Mismatch Error — site CRASHES on load.

Fix: Use custom useStore hook in ALL components that render persisted store data:

```
const useStore = <T, F>(  
  store: (callback: (state: T) => unknown) => unknown,  
  callback: (state: T) => F  
) => {  
  const result = store(callback) as F;  
  const [data, setData] = useState<F>();  
  useEffect(() => { setData(result); }, [result]);  
  return data;  
};
```

Usage: `const cart = useStore(useCartStore, (state) => state.items);`

This delays rendering persisted state until component mounts on client.

### 24.2 Razorpay Webhook Idempotency (Affects Phase 12)

Problem: Razorpay sends the same webhook event multiple times due to network retries.

Result: Inventory decremented twice — phantom stock loss.

Fix: In `/api/razorpay/webhook` handler:

1. Extract `razorpay_payment_id` from event payload
2. `SELECT * FROM orders WHERE idempotency_key = razorpay_payment_id`
3. If found → return 200 immediately (already processed)
4. If not found → process normally (set `idempotency_key` BEFORE downstream actions)

## 24.3 Shiprocket NDR Management (Affects Phase 13)

Problem: COD order returns (RTO) in India are 15–30% of orders if addresses are wrong. Without NDR tracking, these pile up invisibly — lost revenue and poor service rating.

Fix: Shiprocket webhook sends NDR events → update orders.ndr\_status.  
Admin panel must have dedicated NDR tab (see Section 13.2 for full spec).  
Without this, Samorah will lose significant revenue to unmanaged returns.

## 24.4 Abandoned Cart Trigger (Affects Phase 14)

Problem: Next.js has no built-in background timers. You cannot set `setTimeout(2 hours)` on a server.

If you try to track abandonment from the frontend, users closing the browser kills all timers.

Fix: Use Supabase `pg_cron` (server-side cron jobs in PostgreSQL):

1. Cart table has `last_activity_at` TIMESTAMP field

2. `pg_cron` job runs every hour:

```
SELECT * FROM cart WHERE last_activity_at < NOW() - INTERVAL '1 hour'
AND NOT EXISTS (SELECT 1 FROM orders WHERE user_id = cart.user_id AND
created_at > cart.last_activity_at)
```

3. For each matching cart → call Resend API → send abandoned cart email

4. `pg_cron` is enabled in Supabase via: `CREATE EXTENSION pg_cron;`

## 24.5 GST Calculation Logic (Affects Phase 12 + Invoice)

Problem: Indian GST depends on whether the order is intra-state or inter-state.

- If customer state = seller state: charge CGST (half rate) + SGST (half rate)
- If customer state != seller state: charge IGST (full rate)

For candles (HSN 3406, 12% GST):

- Intra-state: CGST = 6% + SGST = 6%
- Inter-state: IGST = 12%

Fix: In checkout API, compare customer address state with `settings.store_state`.

Calculate `cgst_amount`, `sgst_amount` OR `igst_amount` accordingly.

Store all three on orders table — zero out the unused fields.

Tax Invoice PDF must display the correct split per Indian GST law.



## 25. Indian Legal & Tax Compliance

### 25.1 GST Requirements

As a registered GST business, Samorah must issue legally valid Tax Invoices. A regular 'receipt' or 'order confirmation' is NOT sufficient under Indian law. Non-compliance attracts penalties from the GST department.

### 25.2 Mandatory Tax Invoice Fields

| Field                | Requirement                                                    |
|----------------------|----------------------------------------------------------------|
| Invoice Title        | Must say 'TAX INVOICE' — not 'Receipt' or 'Order Confirmation' |
| Invoice Number       | Sequential unique number — e.g. SAM/2025-26/0001               |
| Invoice Date         | Date of supply (payment date)                                  |
| Seller GSTIN         | Samorah's GSTIN — mandatory                                    |
| Seller Address       | Registered business address                                    |
| Buyer Name + Address | Customer's delivery address                                    |
| HSN Code per item    | Each line item must show HSN code (e.g. 3406 for candles)      |
| Taxable Value        | Price before GST for each item                                 |
| CGST / SGST or IGST  | Tax amount per item and total — intra vs inter-state split     |
| Total with GST       | Total amount including all taxes                               |
| Payment Method       | UPI / Card / Net Banking / COD                                 |

### 25.3 Other Legal Requirements

| Requirement        | Implementation                                                                     |
|--------------------|------------------------------------------------------------------------------------|
| Privacy Policy     | DPDP-compliant — data collected, how used, user rights, contact — linked in footer |
| Terms & Conditions | Indian governing law clause, dispute resolution, product liability                 |
| Return Policy      | Eligibility, non-returnable items, damaged goods process, refund timelines         |
| Shipping Policy    | Delivery timelines, free shipping conditions, COD availability                     |
| Cookie Consent     | Cookie consent banner — DPDP Act requires explicit consent for analytics cookies   |

|                |                                                                             |
|----------------|-----------------------------------------------------------------------------|
| Data Retention | Order/invoice data: 7 years (GST law)   User data: delete on request (DPDP) |
| GSTIN Display  | Seller's GSTIN displayed in footer and on all invoices                      |

## 26. User Scenario Document (USD) — 40 Real-World Flows

WHY THIS EXISTS: Claude should build according to user journeys, not pages.

Each scenario below is a complete workflow that drives what code needs to be built.

Reference these scenarios when prompting Claude for each phase.

A feature that isn't in any scenario should be questioned — is it really needed?

### 26.1 Customer Flows — 15 Scenarios

| #  | Scenario                     | Complete Flow                                                                                                                                                                                             |
|----|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C1 | Guest first visit            | Google search → Homepage (hero, chapters) → Collection page → Product page (fragrance notes) → Add to cart (mini cart opens) → Guest checkout → Razorpay UPI → Order confirmed → Email sent               |
| C2 | Registered user full journey | Login (Google) → Wishlist a product → Browse → Cart → Apply coupon WELCOME20 → Saved address → Shiprocket rate displayed → Razorpay card → Order confirmed → Track in /account/orders                     |
| C3 | Bundle purchase              | Homepage → Bundle Builder page → Filter by Ceramic → Select 3 candles → Composition tray fills → Savings displayed → Add to cart → Checkout → Pay → Bundle shipped                                        |
| C4 | Coupon application           | Cart → Coupon field → Enter DIWALI25 → API validates → Discount applied → Total updates → Checkout → Pay                                                                                                  |
| C5 | Gift card purchase & use     | Go to /gift-cards → Select ₹1000 → Pay via Razorpay → Gift card email sent to recipient → Recipient uses code at cart → Balance deducted → Remaining balance preserved                                    |
| C6 | Gift order placement         | Cart → 'This is a gift' checkbox → Gift note field → Recipient name → Occasion: Birthday → Checkout → Packing slip shows gift note, hides pricing                                                         |
| C7 | Loyalty points redemption    | Login → Account → Loyalty section → 500 points balance → Cart → 'Redeem 200 points' → ₹20 discount → Checkout → Points deducted → Remaining balance: 300                                                  |
| C8 | Referral program             | Login → Account → Copy referral link → Share via WhatsApp → Friend clicks link → ref= param captured → Friend registers → First purchase → Referrer gets ₹100 coupon email → Friend gets ₹100 off applied |
| C9 | Payment failure + retry      | Checkout → Razorpay overlay → UPI timeout → Payment failed → payment.failed webhook → Stock reservation                                                                                                   |

|     |                               |                                                                                                                                                                                                                             |
|-----|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |                               | released → Failure email sent → Customer shown 'Retry Payment' → New Razorpay order created → Retry succeeds                                                                                                                |
| C10 | COD order                     | Checkout → Delivery: COD available for pincode → Select COD → Review → Place Order → Order status: confirmed → Shiprocket COD order created → Delivered → COD collected → payment_status: paid                              |
| C11 | Abandoned cart recovery       | Add to cart → Close browser → 1hr: Email 1 'Your candles wait' → No action → 24hr: Email 2 → No action → 72hr: Email 3 with 10% off coupon → Customer returns → Coupon applied → Checkout                                   |
| C12 | Order cancellation (pre-ship) | Order placed → Customer wants to cancel → /account/orders/[id] → Cancel Order button (before pickup) → Shiprocket order cancelled → Razorpay refund initiated → Stock restored → Refund email sent                          |
| C13 | Return request (post-deliver) | Delivered → Customer reports broken candle via contact form → Admin opens ticket → Admin initiates Razorpay refund → Stock optionally restocked → Refund email sent with timeline                                           |
| C14 | Product review                | Delivered → 3 days later: Review request email → Click 'Write Review' → Rate 5 stars → Add title + comment + photo → Submit → Pending status → Admin approves → Review visible on product page → 25 loyalty points credited |
| C15 | Out of stock notification     | Product page → Variant out of stock → 'Notify Me' button → Enter email → Stock restored → Auto email: 'Back in stock!' → Customer clicks → Purchases                                                                        |

## 26.2 Admin / Operations Flows — 15 Scenarios

| #  | Scenario                             | Complete Flow                                                                                                                                                                                                                                                                                                     |
|----|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A1 | Add new candle product               | Admin → Products → Add Product → Step 1: name, slug, category Vol.I, price, HSN 3406, GST 12%, weight → Step 2: 3 variants (100g/140g/180g, SKUs auto-generated) → Step 3: Upload 4 images (Cloudinary) + alt texts → Step 4: Top/Heart/Base notes → Step 5: SEO → Step 6: Preview → Publish → Live in <60s (ISR) |
| A2 | Bulk product upload (10–30 products) | Admin → Products → CSV Import → Download template → Fill product data → Upload → System shows validation report (errors highlighted per row) → Fix errors → Re-upload → Preview all 25 imports → Confirm → SKUs auto-generated → All active                                                                       |
| A3 | Change product price                 | Admin → Products → Edit [product] → Change price from ₹599 to ₹649 → Save → Audit log: user_id   action: price_changed   before: {price: 599}   after: {price: 649} → ISR revalidation triggered                                                                                                                  |



|     |                               |                                                                                                                                                                                                                                                                                        |
|-----|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A4  | Process incoming order        | Order notification → Dashboard new order → Open order → Verify payment (Razorpay paid) → Print packing slip → Pack → Update status: Packed → Admin: Create Shiprocket Shipment → Assign Delhivery → AWB generated → Print label → Update: Shipped → 'Shipped' email sent automatically |
| A5  | Handle NDR (failed delivery)  | Shiprocket NDR webhook → orders.ndr_status = delivery_failed → NDR Alert on dashboard → Admin → Orders → NDR Tab → See: SAM-2025-0042   Blue Dart   'Address not found' → Call customer → Correct address → Click 'Instruct Re-attempt' → Shiprocket re-attempts                       |
| A6  | Handle RTO (return to origin) | NDR tab → Order shows 3 failed attempts → Click 'Accept RTO' → Shiprocket marks for return → When returned: Admin confirms receipt → Inventory restocked (if product intact) → Razorpay refund if COD not collected                                                                    |
| A7  | Inventory restock             | Dashboard: Low Stock Alert (SAM-CAN-001-180G = 3 units) → Inventory → Adjust Stock → Add 50 units → Reason: 'New batch received' → inventory_log entry created → Alert cleared                                                                                                         |
| A8  | Initiate customer refund      | Orders → Find order → Verify complaint → Click 'Initiate Refund' → Enter amount (full/partial) → Reason → Confirm → Razorpay refund API called → orders.payment_status = refunded → Refund email sent → Audit log entry                                                                |
| A9  | Create Diwali campaign        | Banners → Edit 'hero' slot → Upload Diwali image → 'Illuminate Your Diwali' headline → 'Shop Diwali Edit' CTA → Save → Homepage hero updates live → Blog: Write Diwali collection story → SEO → Publish → Email: Create Diwali campaign email → Send to newsletter list                |
| A10 | Create and launch coupon      | Admin → Coupons → Create → Code: DIWALI25 → Type: 25% off → Min order: ₹500 → Max uses: 500 → Expiry: Oct 31 → First order only: No → Active → Copy code → Share in email campaign                                                                                                     |
| A11 | Publish blog post             | Admin → Blog → New Post → Title: 'Vol. I: The Dessert Chapter Story' → Rich editor: write content → Insert product cards (Kashmiri Chai, Modak) → Upload cover image → SEO title + description → Tags: vol-1, dessert → Schedule: tomorrow 10am → Published automatically              |
| A12 | Update homepage banner        | Admin → Banners → Click 'hero' slot → Update heading: 'New: Vol. IV — Nature Chapter' → Upload new hero image → CTA: 'Explore Now' → Save → Homepage revalidated → Live in <60s — no developer needed                                                                                  |
| A13 | Manage wholesale inquiry      | Wholesale form submitted → wholesale_customers table entry (status: pending) → Admin reviews → Approves: status = approved, custom_price_tier set → Account created → Wholesale pricing applies at checkout                                                                            |

|     |                           |                                                                                                                                                                                                                |
|-----|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A14 | Monthly performance audit | Admin → Performance Audits → New Audit → Run Lighthouse (paste scores) → Check broken links report → Review low stock → Check email open rates → Check NDR rate → Log all results → Download action items list |
| A15 | Monthly backup            | Admin → Settings → Backup → Export Database → Export Products CSV → Export Orders CSV (for GST records) → Export Customers CSV → Upload to secure cloud storage → Log backup completion date                   |

## 26.3 Marketing & Security Flows — 10 Scenarios

| #  | Scenario                         | Complete Flow                                                                                                                                                                                                                                    |
|----|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M1 | New collection launch            | Pre-launch: Banner Manager teaser hero → Blog draft: collection story → Newsletter teaser → Launch day: Publish products → Publish blog → Update hero banner → Send launch email → Monitor GA4 events (view_item_list surge)                     |
| M2 | Newsletter campaign              | Admin → Newsletter → Select segment (all active subscribers) → Choose template → Customize subject + content → Preview → Send test to admin email → Schedule: next Tuesday 10am → Send → Monitor Resend dashboard for open/click rates           |
| M3 | Festival email campaign (Diwali) | 3 weeks before: Segment newsletter list → Create campaign: subject 'Light up Diwali with Samorah' → Products: festival candles + bundles + gift sets → CTA: 'Shop Diwali Edit' with DIWALI25 coupon → Schedule → Send → Track conversions in GA4 |
| M4 | Birthday campaign automation     | pg_cron daily: SELECT users WHERE birthday = TODAY → For each user: Send birthday email with 100 bonus points → loyalty_transactions entry: type=bonus, points=100 → User receives points notification                                           |
| M5 | AI search (Phase 2)              | Customer types 'cozy warm vanilla candle' → AI embedding (OpenAI) converts to vector → Semantic search against product embeddings → Returns: Kashmiri Chai, Gajar Halwa Delight → Displayed with fragrance notes highlighted                     |
| S1 | Forgot password                  | Login page → Forgot password → Enter email → Reset email sent via Resend → Click link (30-min expiry) → New password form → Password updated → Login → Success                                                                                   |
| S2 | OTP login                        | Login page → OTP tab → Phone number → 6-digit SMS OTP sent → Enter OTP → Verified → JWT issued → Logged in                                                                                                                                       |
| S3 | Role change (Super Admin)        | Super Admin → Customers → Find user → Change role to Editor → Save → Audit log: role_changed   before: customer   after: editor → User now has blog + banner access                                                                              |

|    |                           |                                                                                                                                                                                                                                      |
|----|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| S4 | Duplicate webhook attempt | Razorpay sends payment.captured → System processes: inventory-=1, Shiprocket created, email sent → Razorpay resends same event (network retry) → API checks: idempotency_key exists → Return 200 immediately → Nothing happens twice |
| S5 | Price change audit trail  | Manager changes candle price from ₹599 → ₹699 → Audit log records: user_id   action: price_changed   entity: product   before: {price: 599}   after: {price: 699}   ip_address   timestamp → Super Admin can review anytime          |

## 27. Claude Prompting Strategy — How to Build Phase by Phase

### 27.1 The Three Rules

Rule 1: NEVER give Claude this entire BRD and ask it to 'build the website'.  
Claude will lose context and hallucinate after ~8,000 tokens.

Rule 2: ALWAYS start each phase with a fresh context prompt referencing this BRD.  
Upload the BRD PDF and reference specific section numbers.

Rule 3: NEVER ask Claude to build an entire page. Always ask for one file at a time.  
'Build the ProductGallery component' → then → 'Wire it to Supabase'.

### 27.2 Session Startup Prompt (Use at Start of Every Session)

**Say this exactly at the start of every Claude session:**

"This is the SAMORAH ecommerce project. Platform: Next.js 15 + Supabase + Tailwind + Shadcn + Zustand + Razorpay + Shiprocket + Resend. I have a BRD document. Today we are working on Phase [N]: [Phase Name]. I will give you one task at a time. Do not write code until I give you the specific task."

### 27.3 Phase-by-Phase Prompt Templates

| Phase | Name     | Claude Prompt Template                                                                                                                                                                                                                                                                                      |
|-------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3     | Setup    | 'Referring to SAMORAH BRD Section 3, generate the complete folder structure and install all dependencies. Create: package.json, tsconfig.json, next.config.ts with security headers, and globals.css with all design token CSS variables from Section 4.3.'                                                 |
| 4     | Database | 'Referring to SAMORAH BRD Sections 8.2–8.5, generate the complete Supabase SQL migration file with ALL tables, foreign keys, ENUM types, RLS policies, and indexes. Include the audit_logs, gift_cards, loyalty_transactions, referral_codes, instagram_gallery, homepage_banners tables from Section 8.5.' |
| 5     | Auth     | 'Build the Supabase Auth integration for SAMORAH. Create: 1) AuthProvider component, 2) useAuth hook, 3) Register form with email verification, 4) Login form with Google OAuth and OTP, 5) Next.js middleware for route protection. Reference Section 6 RBAC — protect /admin routes for role: editor+.'   |

|    |             |                                                                                                                                                                                                                                                                                              |
|----|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6  | State       | 'Create all 8 Zustand stores from SAMORAH BRD Section 7. CRITICAL: Implement the hydration-safe useStore hook from Section 24.1 before creating any store with localStorage persistence. Start with useCartStore and useUserStore.'                                                          |
| 12 | Razorpay    | 'Build the Razorpay integration for SAMORAH following Section 12. CRITICAL: The webhook handler at /api/razorpay/webhook must implement idempotency checking as specified in Section 24.2 — check orders.idempotency_key before processing any payment.captured event.'                      |
| 13 | Shipping    | 'Build the Shiprocket integration from SAMORAH BRD Section 13. Include: 1) Rate calculator at checkout, 2) Order creation on payment.captured, 3) AWB generation, 4) NDR webhook handler that updates orders.ndr_status. Also build the admin NDR Tab component (Section 13.2).'             |
| 14 | Email       | 'Build the email system from SAMORAH BRD Section 14. Start with these 5 transactional templates using React Email: 1) Order Confirmed (include Tax Invoice PDF attachment per Section 25.2), 2) Order Shipped, 3) Abandoned Cart Email 1, 4) Welcome, 5) Low Stock Alert. Use Resend API.'   |
| 15 | Admin Panel | 'Build the Admin Panel Dashboard from Section 11.2. Include: Revenue card, Orders card, Inventory Alert card, NDR Alert widget, 30-day revenue chart (Recharts), Recent Orders table (TanStack Table), Low Stock inline table, Quick Actions bar.'                                           |
| 16 | Inventory   | 'Build the Inventory Management module from SAMORAH BRD Section 10. Include: 1) Inventory dashboard screen, 2) Stock adjustment form with audit log entry, 3) Low stock email trigger via Resend, 4) CSV bulk upload with validation preview.'                                               |
| 25 | GST         | 'Implement GST calculation logic from SAMORAH BRD Section 25.4. Build: 1) Function: calculateGST(cartItems, customerState, storeState) that returns cgst/sgst or igst amounts, 2) Update checkout API to store tax amounts, 3) Tax Invoice PDF generator with all fields from Section 25.2.' |

## 27.4 One File at a Time — Product Page Example

| Step | Prompt           | Output                                                                                                                                                                                                 |
|------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Gallery first    | 'Build only the ProductGallery component — image thumbnails, main image, zoom on hover, mobile swipe. No data fetching yet. Use the 1:1 image ratio from Section 4.4.'                                 |
| 2    | Variant selector | 'Build the VariantSelector component — vessel type buttons (Glass/Ceramic/Terracotta), size buttons (100g/140g/180g), price update on selection, out-of-stock disabling. Props: variants[], onSelect.' |

|   |              |                                                                                                                                                                                                                                                             |
|---|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 | Wire to data | 'Now wire ProductGallery and VariantSelector to Supabase. Create getProduct(slug) service function that fetches products JOIN variants JOIN product_images JOIN fragrance_notes. Make this a server component using Next.js fetch with ISR revalidate: 60.' |
| 4 | Add to cart  | 'Add the AddToCart button to the product page. On click: call useCartStore.addItem() with selected variant. Also add AddToWishlist button using useWishlistStore. Remember: use the hydration-safe useStore hook from Section 24.1.'                        |
| 5 | GA4 events   | 'Add GA4 event tracking to the product page following Section 18.1: fire view_item on page load, add_to_cart on cart add, add_to_wishlist on wishlist add. Parameters must include item_id (SKU), item_name, price, currency: INR.'                         |

## 27.5 Three Supporting Documents (Future)

| Document                       | Purpose                                                                | When to Create                                                    |
|--------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------|
| PRD (Product Requirements Doc) | Features with acceptance criteria — 'when is this done?'               | Before Phase 8 (Homepage) — define done criteria for each feature |
| USD (User Scenario Doc)        | 40–50 user workflows (Section 26 is the foundation)                    | Before Phase 11 (Cart) — ensure all payment flows are covered     |
| SAD (System Architecture Doc)  | Database ERD, API endpoints list, security diagram, deployment diagram | Before Phase 18 (Deployment) — for handoff to any developer       |

## 28. Deployment, Timeline & Cost

### 28.1 Environment Architecture

| Environment | URL                | Branch                                |
|-------------|--------------------|---------------------------------------|
| Development | localhost:3000     | feature/* branches — hot reload       |
| Staging     | staging.samorah.in | develop branch — auto-deploy on merge |
| Production  | samorah.in         | main branch — zero-downtime deploy    |

### 28.2 Updated Phase Timeline (27 Phases)

| Ph | Phase            | Key Deliverables                                               | Week   | Depends |
|----|------------------|----------------------------------------------------------------|--------|---------|
| 1  | Branding         | Typography, tokens, spacing, animation system                  | Wk 1   | —       |
| 2  | Design System    | Buttons, inputs, cards, drawer, modal, toast                   | Wk 1–2 | Ph 1    |
| 3  | Folder Setup     | Next.js 15, TypeScript, Tailwind, Supabase client, pg_cron     | Wk 2   | Ph 2    |
| 4  | Database         | All 20+ tables, RLS policies, GST fields, audit_logs           | Wk 2–3 | Ph 3    |
| 5  | Authentication   | Register, login, OTP, Google OAuth, RBAC middleware            | Wk 3   | Ph 4    |
| 6  | State (Zustand)  | 8 stores + hydration-safe useStore hook                        | Wk 3–4 | Ph 5    |
| 7  | Product Engine   | Products, variants, SKU system, categories, collections        | Wk 4–5 | Ph 4    |
| 8  | Homepage         | 12 sections + Banner Manager integration                       | Wk 5   | Ph 7    |
| 9  | Collection Pages | Editorial layout, product grid, filters                        | Wk 5–6 | Ph 7    |
| 10 | Product Page     | 12 sections including reviews, fragrance notes, GA4 events     | Wk 6   | Ph 7    |
| 11 | Cart + Checkout  | Cart with gift features, GST-aware checkout, coupon, gift card | Wk 7   | Ph 6    |
| 12 | Razorpay         | Order creation, webhook + idempotency, GST calculation         | Wk 7–8 | Ph 11   |
| 13 | Shiprocket       | Order sync, AWB, NDR webhook, NDR admin tab                    | Wk 8   | Ph 12   |

|            |                    |                                                                        |                  |       |
|------------|--------------------|------------------------------------------------------------------------|------------------|-------|
| 14         | Email System       | 15 templates + abandoned cart (pg_cron) + festival sequences           | Wk 8–9           | Ph 12 |
| 15         | Admin Panel        | Dashboard, products, orders (NDR tab), inventory, coupons, reviews     | Wk 9–11          | Ph 7  |
| 16         | Inventory Mgmt     | SKUs, stock, alerts, CSV import, audit trail                           | Wk 11            | Ph 15 |
| 17         | Loyalty + Referral | Points system, tiers, referral codes, gift cards, account UI           | Wk 11–12         | Ph 5  |
| 18         | Banner Manager     | Homepage Banner Manager + Instagram Gallery Admin                      | Wk 12            | Ph 8  |
| 19         | GST + Tax Invoice  | GST calculation, CGST/SGST/IGST, PDF Tax Invoice generation            | Wk 12            | Ph 12 |
| 20         | SEO + Blog         | Dynamic metadata, sitemap, schema, Sanity CMS blog                     | Wk 12–13         | Ph 8  |
| 21         | Event Tracking     | All GA4 events, Meta Pixel events, Microsoft Clarity                   | Wk 13            | Ph 8  |
| 22         | Review System      | Moderation, images, helpful votes, admin reply                         | Wk 13            | Ph 10 |
| 23         | Backup + Audit     | Audit logs, backup admin screen, export CSVs                           | Wk 13            | Ph 15 |
| 24         | Security + Testing | Rate limiting, CSRF, Jest unit tests, Playwright E2E                   | Wk 13–14         | Ph 23 |
| 25         | Deployment         | Vercel production, CI/CD, domain DNS, monitoring                       | Wk 14            | Ph 24 |
| 26         | Product Upload     | Upload 10–30 initial products with all metadata                        | Wk 14            | Ph 25 |
| 27         | Launch QA          | Full device testing, payment test, Shiprocket test, GST invoice review | Wk 14–15         | Ph 26 |
| <b>MVP</b> | <b>TOTAL</b>       | Complete luxury ecommerce with admin, inventory, loyalty, GST          | <b>14–16 wks</b> | —     |

## 28.3 Monthly Infrastructure Cost

| Service | Plan at Launch | Cost/Month | Upgrade Trigger       |
|---------|----------------|------------|-----------------------|
| Vercel  | Hobby (free)   | ₹0         | When >100GB bandwidth |



|                |                     |            |                               |
|----------------|---------------------|------------|-------------------------------|
| Supabase       | Free tier           | ₹0         | When >500MB DB or 2GB storage |
| Cloudinary     | Free tier           | ₹0         | When >25GB storage            |
| Resend         | Free tier           | ₹0         | When >3,000 emails/month      |
| Razorpay       | Pay-per-transaction | 2% per txn | No monthly fee ever           |
| Shiprocket     | Pay-per-shipment    | Per order  | No monthly fee                |
| GoDaddy domain | Annual              | ~₹65/mo    | Annual renewal                |
| <b>TOTAL</b>   | —                   | ₹65–500/mo | vs ₹2,500–8,000 on Shopify    |

